

Date	15/07/2026
Team ID	xxxxxxx
Project Name	Smart Lender (Loan Approval Prediction)
Maximum Marks	2 Marks

Data Flow Diagram Template

Data Flow Diagram (DFD) details and process descriptions.

Symbol Name	Description	Example in Smart Lender
External Entity	Interacts with the system.	Loan Applicant, Bank Officer
Process	Transforms input data.	Form Input, Scaling, RandomForest Inference, HTML Render
Data Store	Stores persistent information.	model.pkl, scaler.pkl, train.csv
Data Flow	Movement of data.	Applicant Details (Form -> Flask), Scaled Vector (Flask -> Model)

Data Flow Steps:

1. The Applicant enters details through the Web Form. This forms the raw input data.
2. The data flows into the Flask Server (app.py) where a mapping process translates categorical text to numerical values.
3. The server sends the features to StandardScaler (scaler.pkl), which outputs scaled features.
4. The scaled features flow into the Random Forest Model (model.pkl), which returns a prediction (0 or 1).
5. Flask translates the output to a text message and renders it on [submit.html](file:///C:/Users/smyas/Downloads/SmartLender/SmartLender/5.Project%20Development%20Phase/templates/submit.html) to the Applicant.